

K. PUNITH RAJ

Aspiring AI/ML Engineer | Generative AI · LLMs · RAG · Agentic Systems | Python · PyTorch · LangChain/LangGraph

Bengaluru, Karnataka, India | punithraj कुमार602@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Aspiring AI/ML Engineer with hands-on, end-to-end project experience across Machine Learning, Deep Learning, and Generative AI — including RAG pipelines, multi-agent LLM systems, and LLM evaluation frameworks. Proficient in Python, PyTorch, and LangChain/LangGraph for model and agent development, with FastAPI for serving and Docker for deployment. Every project is built and deployed for real use rather than left as a notebook. Seeking Fresher/Junior AI/ML Engineer, Generative AI Engineer, or LLM Engineer roles and internships in Bengaluru or remote.

SKILLS

- **Programming & Data:** Python, SQL, Pandas, NumPy, Data Preprocessing & Analysis
- **ML / Deep Learning:** Machine Learning, Deep Learning, Neural Networks, CNN, NLP, Computer Vision
- **Generative AI / LLM:** LLMs, RAG, Multi-Agent Systems, Prompt Engineering, Evaluation & Guardrails
- **Frameworks:** PyTorch, TensorFlow, Scikit-Learn, LangChain, LangGraph, FastAPI, Streamlit
- **Deployment / Tools:** Docker, Git, GitHub, Vector Databases, REST APIs, Model Deployment

PROJECTS

Multimodal RAG Pipeline for Scanned, Tabular & Visual Documents — *OCR + Vision-LLM + Retrieval-Augmented Generation*

[GitHub Repository](#) | [Live Demo](#)

- Built and deployed a RAG pipeline that unifies native text, OCR-extracted scanned pages, and structured tables into a single searchable index, with an integrated vision-LLM to answer questions about embedded charts and images.
- Validated page-type classification and table detection through automated testing across text, scanned, tabular, and image document types, with an average parse time of ~30ms per document.

RAG Evaluation & Guardrails Framework — *Automated faithfulness/relevancy scoring with safety guardrails*

[GitHub Repository](#) | [Live Demo](#)

- Designed and deployed an evaluation framework that scores RAG answers for faithfulness and relevancy, paired with guardrails that flag unsafe, PII-containing, or ungrounded outputs before they reach users.
- Validated the framework against a hand-labeled test set of faithful, hallucinated, off-topic, and unsafe/PII cases, correctly detecting all 25 labeled cases across the evaluator and guardrail layers.

Multi-Agent Code Review System — *Parallel LLM agents for style, bug, and test-coverage review*

[GitHub Repository](#) | [Live Demo](#)

- Built a multi-agent system where three specialized LLM agents (style, bug-finder, test-coverage) independently review the same code change in parallel and are consolidated into a single report, exposed via both a FastAPI endpoint for CI/webhook use and an interactive UI.
- Validated end-to-end orchestration and real pytest execution across a multi-file test batch, with every file receiving complete, non-crashing output from all three agents.

Self-Improving Agent with Reflection & Long-Term Memory — *Draft → self-critique → revise loop with persistent vector memory*

[GitHub Repository](#) | [Live Demo](#)

- Built an agent that drafts an answer, critiques it against a rubric, and revises when the critique finds issues, then persists every interaction (including corrected mistakes) to long-term vector memory to inform future responses.
- Verified memory-context formatting, JSON-response parsing, and graceful fallback under simulated API failure through automated reliability testing, with all checks passing.

EDUCATION

Bachelor of Computer Applications (BCA)

Mahatma Gandhi University (MGU), Kottayam | June 2020 – June 2023

CERTIFICATIONS & LANGUAGES

- **Certification:** AI & Machine Learning — Valaxy Technologies (May 2026)
- **Languages:** English (Professional Working) · Kannada (Full Professional) · Telugu (Full Professional)